

Build advanced and highly reliable industrial Ethernet communication system



# ISM5310D PoE Managed Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

# 1. Product Overview

The ISM5310D PoE managed industrial Ethernet switch is specifically designed to meet the demands of harsh industrial environments such as power, transportation, and industrial control applications.

## Product Features:

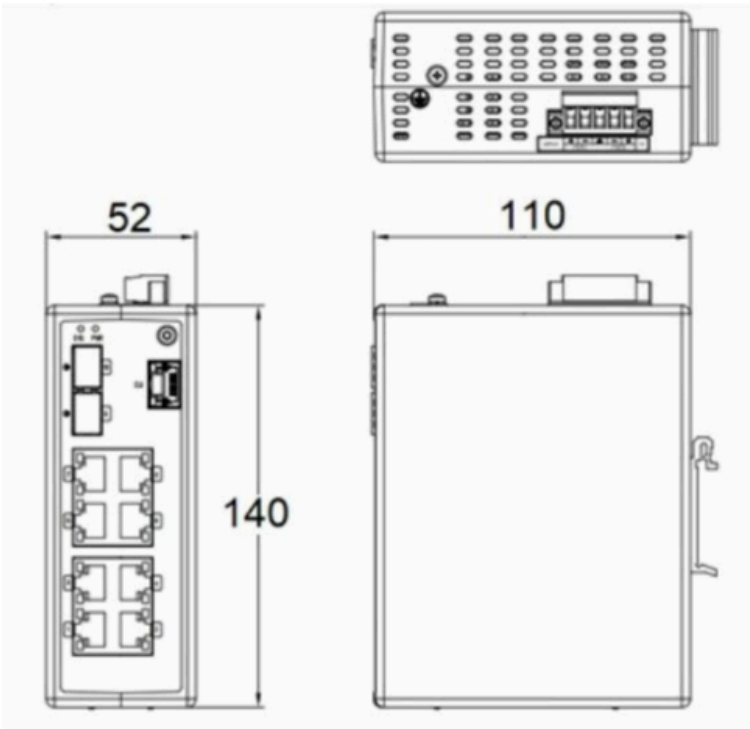
- **Reliability & Resilience:** Fanless design, IP30 protection rating, dust and dirt resistance, and wide temperature operation.
- **Redundancy Protocols:** Supports STP/RSTP/MSTP ring redundancy protocols, offering flexible choices for intricate industrial networks.
- **Network Management:** Supports SNMP for integrated network management and RMON for effective network monitoring and fault prediction.
- **Enhanced Security & Control:** Port-based VLAN, IEEE 802.1Q VLAN, GVRP, IGMP Snooping, and QoS to process critical data.
- **PoE Capabilities:** Supports IEEE 802.3af/at PSE, providing up to 30W output power per port.

## Key Technical Specifications

| Parameter             | Specification                                                      |
|-----------------------|--------------------------------------------------------------------|
| Type                  | Layer 2 Managed Industrial Ethernet Switch                         |
| Ports                 | 2 x 100/1000/2500BaseX SFP + 8 x 10/100/1000BaseT PoE (802.3af/at) |
| Switching Performance | 44 Gbps backplane bandwidth; 8K MAC table                          |
| Dimensions / Weight   | 52 mm x 140 mm x 110 mm; 0.7 kg (1.54 lb)                          |
| Power                 | 48~57 VDC dual redundant input                                     |
| Environment           | -40 °C ~ +75 °C (-40 °F ~ +167 °F) operating; IP30                 |

| Parameter      | Specification                                                     |
|----------------|-------------------------------------------------------------------|
| Management     | Web, CLI, SNMPv1/v2c/v3, RMON; STP/RSTP/MSTP, IGMP Snooping, VLAN |
| Certifications | CE, FCC                                                           |

## 2. Product Dimensions



ISM5310D

- Note:
- 1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
  - 2. All dimensions are approximate and **for reference only**.
  - 3. The dimensions shown in the figure **shall not be used for production or processing**.
  - 4. Dimensions must comply with part and manufacturing tolerance requirements.
  - 5. Dimensions are subject to change without notice.

## 3. Technical Specifications

## 3.1 Protocol Compliance List

| Category/Parameter    | Specification                                    |
|-----------------------|--------------------------------------------------|
| <b>IEEE Standards</b> |                                                  |
| IEEE 802.3            | CSMA/CD method and physical Layer specifications |
| IEEE 802.1p           | Priority Queuing                                 |
| IEEE 802.1q           | VLAN tagging                                     |
| IEEE 802.1d           | Spanning Tree Algorithm                          |
| IEEE 802.1w           | Rapid Spanning Tree                              |
| IEEE 802.1s           | Multiple Spanning Tree                           |
| IEEE 802.3ac          | VLAN Tagging                                     |
| IEEE 802.1x           | Authentication                                   |
| IEEE 802.3ad          | Link Aggregation                                 |
| IEEE 802.3x           | Flow Control                                     |
| IEEE 802.3            | Ethernet                                         |
| IEEE 802.3u           | Fast Ethernet                                    |
| IEEE 802.3z           | Gigabit Ethernet                                 |
| IEEE 802              | Networks                                         |
| <b>RFC Standards</b>  |                                                  |
| RFC 768               | UDP                                              |
| RFC 791               | IP                                               |
| RFC 792               | ICMP                                             |
| RFC 793               | TCP                                              |

| Category/Parameter | Specification                                       |
|--------------------|-----------------------------------------------------|
| RFC 826            | ARP                                                 |
| RFC 854            | Telnet Client & Server                              |
| RFC 862            | Echo Protocol                                       |
| RFC 863            | Discard Protocol                                    |
| RFC 867            | Daytime Protocol                                    |
| RFC 868            | Time Protocol                                       |
| RFC 1059, 1119     | NTPv1/2                                             |
| RFC 1191           | Path MTU Discovery                                  |
| RFC 1403           | BGP OSPF Interaction                                |
| RFC 1542           | Bootstrap Extensions & DHCP                         |
| RFC 1994           | PPP Challenge Handshake Authentication Protocol     |
| RFC 2068           | HTTP                                                |
| RFC 213            | DHCP Server                                         |
| RFC 2138           | RADIUS                                              |
| RFC 2139           | RADIUS Accounting                                   |
| RFC 2236           | IGMPv2                                              |
| RFC 2474           | DiffServ Precedence                                 |
| RFC 2597           | DiffServ Assured Forwarding                         |
| RFC 2598           | DiffServ Expedited Forwarding                       |
| RFC 2865           | Remote Authentication Dial In User Service (RADIUS) |
| RFC 3046           | DHCP Relay Agent Information Option                 |

| Category/Parameter     | Specification                     |
|------------------------|-----------------------------------|
| RFC 3222               | Forwarding Information Base (FIB) |
| <b>Other Protocols</b> |                                   |
| GMRP                   | GARP                              |
| GVRP                   | GARP                              |
| SSH2                   | Secure Shell 2                    |
| IGMP                   | Snooping                          |
| SNMPv3                 | Supported                         |

## 3.2 Hardware Specifications

| Category/Parameter          | Specification                           |
|-----------------------------|-----------------------------------------|
| <b>Physical Performance</b> |                                         |
| Enclosure                   | Fully enclosed seamless metal enclosure |
| Dimensions (W × D × H)      | 52 mm × 140 mm × 110 mm                 |
| Weight                      | 0.7 kg (1.54 lb)                        |
| Mounting Method             | DIN-rail mounting                       |
| Cooling Method              | Fanless cooling                         |
| Ingress Protection          | IP30                                    |
| Storage Temperature         | -40 °C ~ +85 °C (-40 °F ~ +185 °F)      |
| Operating Temperature       | -40 °C ~ +75 °C (-40 °F ~ +167 °F)      |
| Humidity                    | 5 ~ 95% (non-condensing)                |

| Category/Parameter                     | Specification                                     |
|----------------------------------------|---------------------------------------------------|
| <b>Hardware Performance</b>            |                                                   |
| Backplane Bandwidth                    | 44 Gbps                                           |
| Transmission Mode                      | Parallel Storage Forwarding                       |
| MAC Table Size                         | 8K                                                |
| RAM                                    | 128 MB                                            |
| Packet Buffer Size                     | 4 Mbits                                           |
| Exchange Rate                          | 148,800 pps/100M ports; 1,488,000 pps/1000M ports |
| <b>Power Parameters</b>                |                                                   |
| Input Voltage                          | 48~57 VDC dual redundant input                    |
| Overload Current Protection            | Supported                                         |
| Reverse Polarity Protection            | Supported                                         |
| <b>PoE</b>                             |                                                   |
| PoE 10/100/1000 BaseT Ports            | 8 ports                                           |
| Maximum Power                          | 15.4W (IEEE 802.3af); 30W (IEEE 802.3at)          |
| <b>Electromagnetic Characteristics</b> |                                                   |
| EMI                                    | FCC 47 CFR Part 15 Class A; EN55022 Class A       |

| Category/Parameter                | Specification                                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| EMS                               | IEC(EN)61000-4-2, Class 4<br>IEC(EN)61000-4-4, Class 4<br>IEC(EN)61000-4-5, Class 4<br>IEC(EN)61000-4-11, Class 4<br>IEC(EN)61000-4-12, Class 4 |
| <b>Mechanical Characteristics</b> |                                                                                                                                                 |
| Shock                             | IEC60068-2-27                                                                                                                                   |
| Freefall                          | IEC60068-2-31                                                                                                                                   |
| Vibration                         | IEC60068-2-6                                                                                                                                    |
| <b>Certifications</b>             |                                                                                                                                                 |
| Certifications                    | CE, FCC                                                                                                                                         |
| <b>Quality Assurance</b>          |                                                                                                                                                 |
| Warranty Period                   | 5 years                                                                                                                                         |
| MTBF                              | 35 years                                                                                                                                        |

## 4. Software Specifications

| Category/Parameter        | Specification                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------|
| <b>Software Functions</b> |                                                                                                |
| Redundancy                | STP, MSTP, RSTP, Port Trunking                                                                 |
| Management Mode           | Browser, Serial Port, STD-17 MIB-II, STD-58 SMIV2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1 |
| Time Synchronization      | NTP                                                                                            |



| Category/Parameter | Specification                                             |
|--------------------|-----------------------------------------------------------|
| Diagnostic Mode    | Indicator light, Journal File, RMON, Port Mirroring, TRAP |

## 5. Ordering Guide

| Model                   | Description                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISM5310D-P-2GSFP-8GT-48 | 10-port Layer2 managed Industrial Switch. 2 * 100/1000/2500BaseX SFP Ports (SFP module not included), 8 * 802.3af/at PoE 10/100/1000BaseT Ports. 1 Management Serial CLI Port, IP30 Protection Class, Operating Temperature from -40 °C to +75 °C (-40 °F to +167 °F). Dual 48-57VDC Power Inputs. |

## 6. Contact Us

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